

Emergency Green Corridor

Warning the junction constable before the siren arrives.

Nandish · green-corridor-2026.web.app

60 %+

of Karnataka 108 cardiac, stroke, respiratory calls missed the 10-minute target

Source: CAG audit of Karnataka 108 EMS, 2014–19

Team details

A solo entry, built end to end by one contributor.

TEAM NAME

Green Corridor

TEAM LEADER

Nandish

CONTRIBUTORS

Sole contributor

PROBLEM STATEMENT

Sustainability & Social Impact: emergency vehicles stuck at red signals while the junction constable is never told.

Brief about the idea

Warn the junction constable early, sized to the queue

≈ 7.9 min

saved for the critical ambulance

4 vs 1 min

warning at a 400 m queue vs a
100 m queue

16.5 min

three-vehicle total, simulated
baseline

Scripted scenario and synthetic voices, not field results.

Opportunities (how it differs / how it solves / USP)

Not a fixed radius, and not waiting for new hardware



HOW IT DIFFERS

**Warning sized to
the queue**

vs fixed-radius alerts



HOW IT SOLVES

**Queue cleared
before the siren**

vs manual green corridors



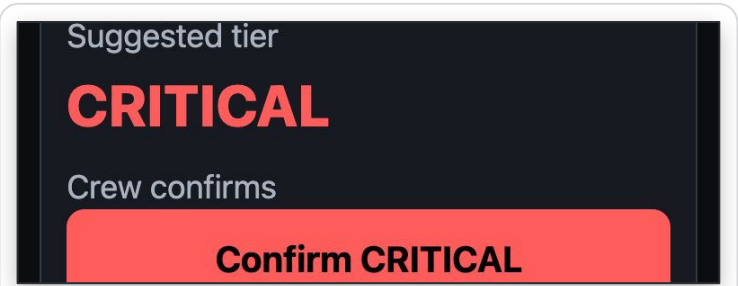
USP

**Rules decide,
Gemini explains;
no hardware**

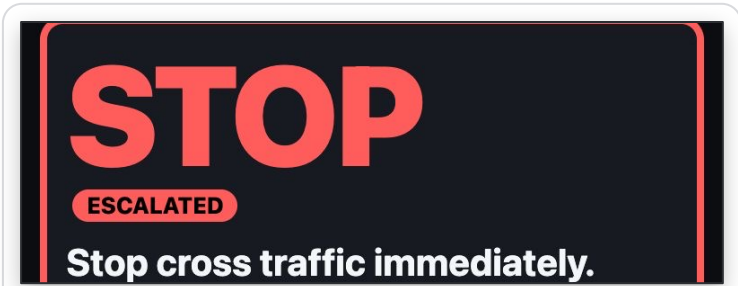
vs IR/GPS preemption

List of features

Six screens and the engines behind them



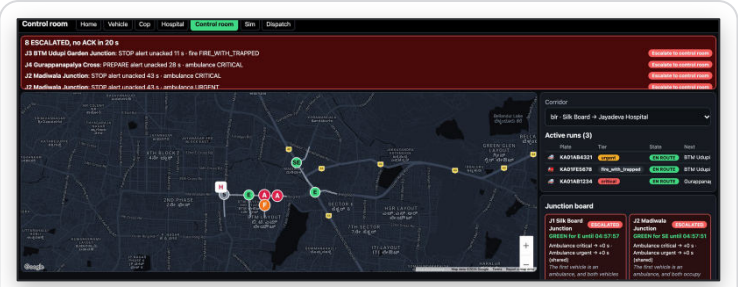
Voice triage



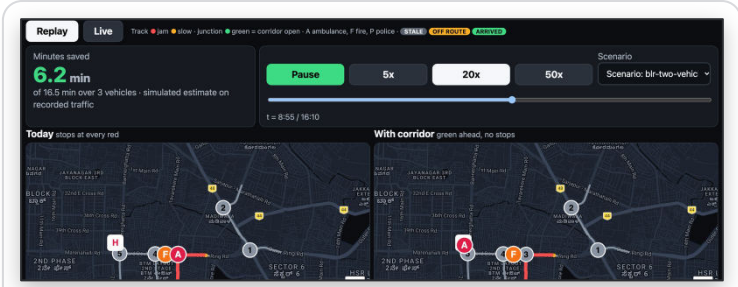
Spoken cop alert



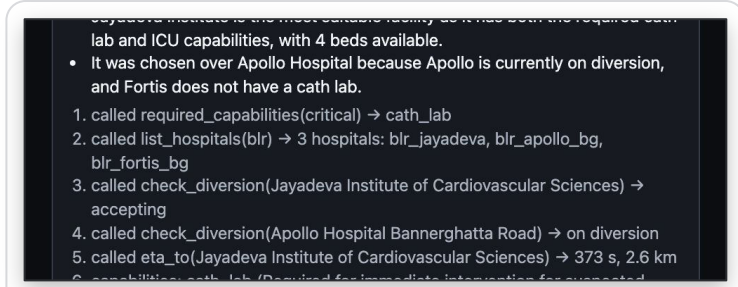
Hospital live brief



Control room board



With-vs-without replay



Routing agent trace

Lead time

Sequencing

Cop voice

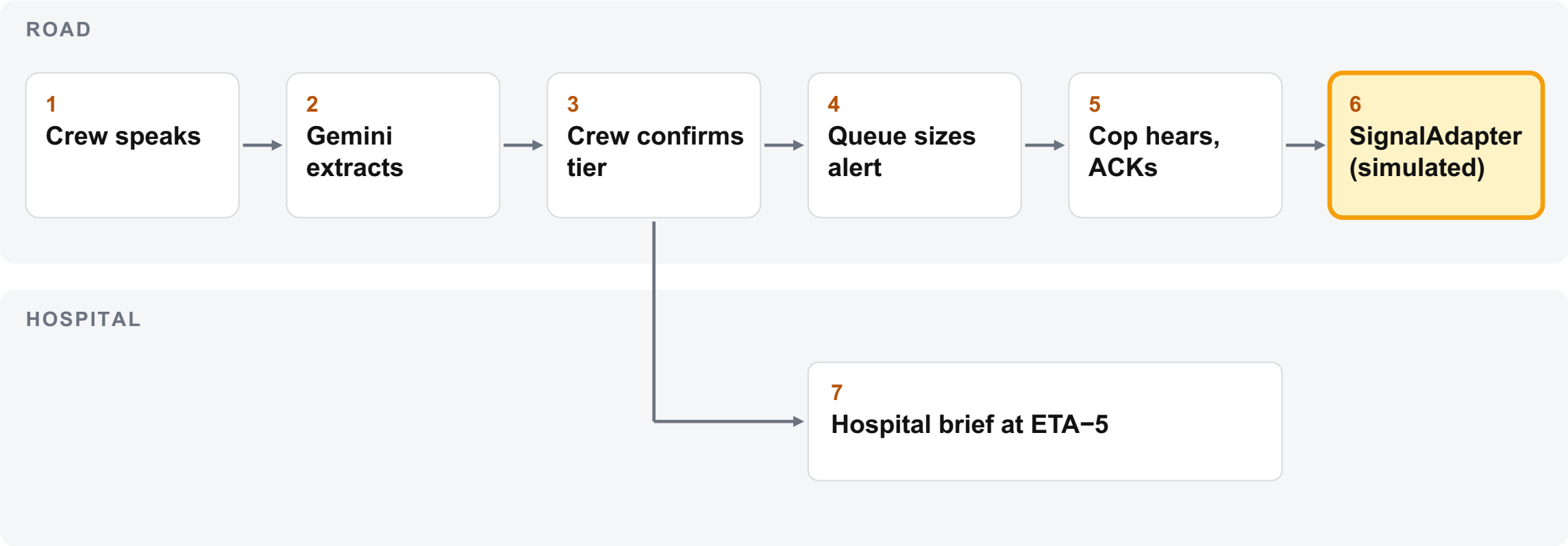
ATMIST

After-action

Eval harness

Process flow

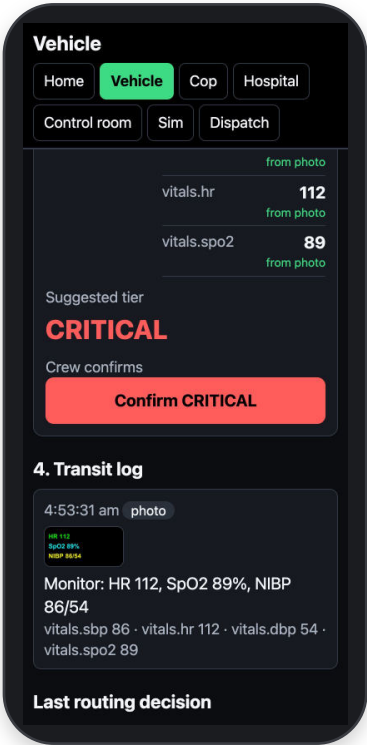
From the crew’s words to the constable’s phone



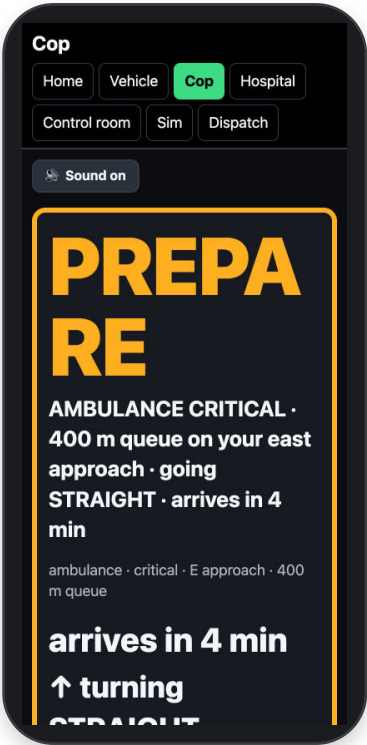
Amber seam: simulated today.

Wireframes

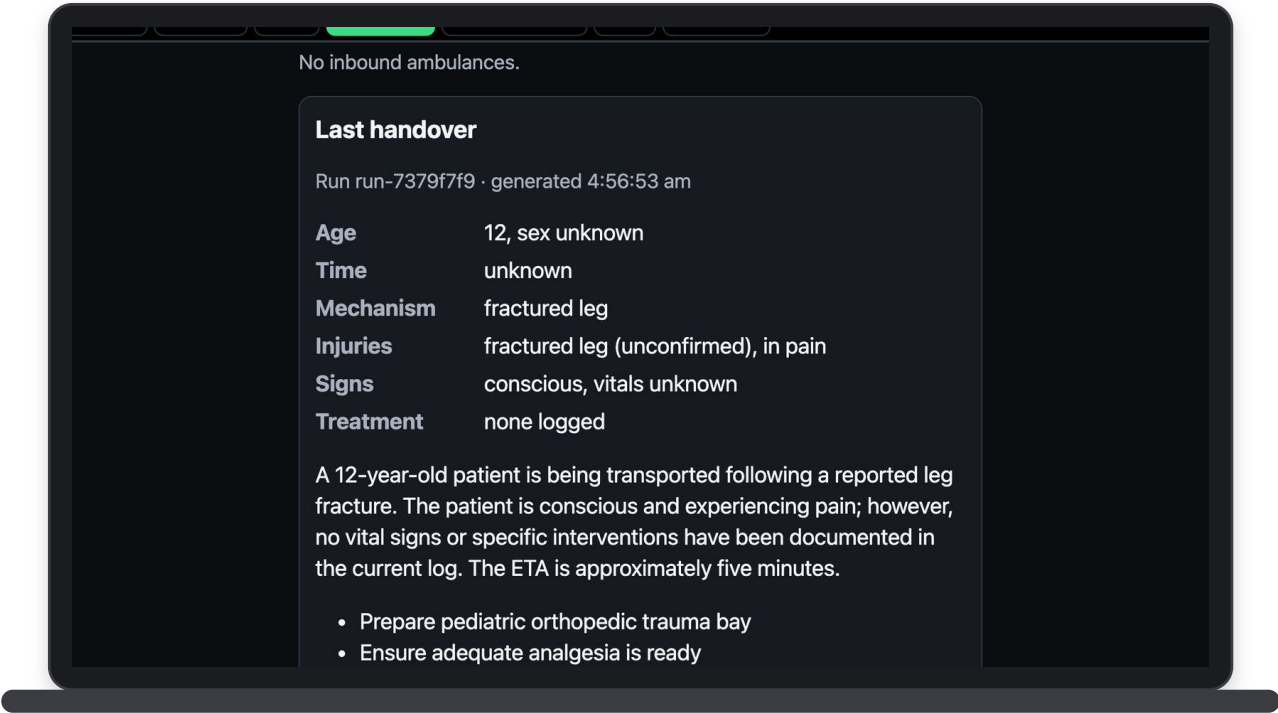
Real screens from the running prototype



Voice triage



PREPARE alert

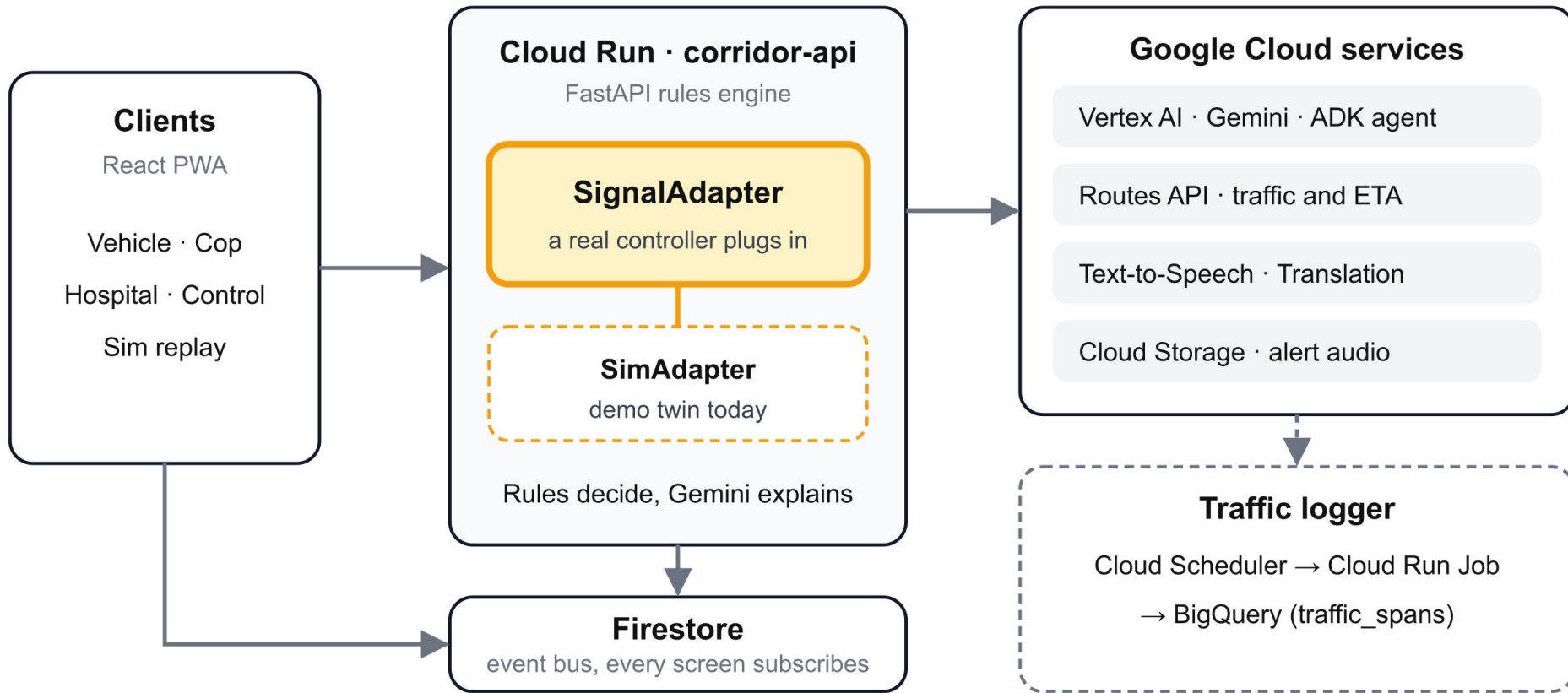


Hospital handover

Live rehearsal captures; synthetic patients.

Architecture

Rules decide; Google Cloud services extract and explain



Rules engine on Cloud Run, Firestore event bus; the amber SignalAdapter seam is where a real controller plugs in.

Technologies (Google Cloud)

A Google Cloud stack, from model to delivery

● AI & language	Gemini on Vertex AI	Google ADK	Cloud Translation
● Maps & speech	Routes API	Text-to-Speech	Cloud Storage
● Run & data	Cloud Run	Firestore	BigQuery
● Delivery	Firebase Hosting	Cloud Build	Secret Manager

BigQuery ML retrained 8 Oct on 2,173 traffic readings, 5–8 Oct, mostly peak hours; 88% still zero jam; baseline, not deployed

Estimated implementation cost

About ₹155 per vehicle-hour, almost all of it Routes

≈ ₹155

per vehicle-hour (estimate)

≈ ₹50

per 20-min run

≈ ₹5,000

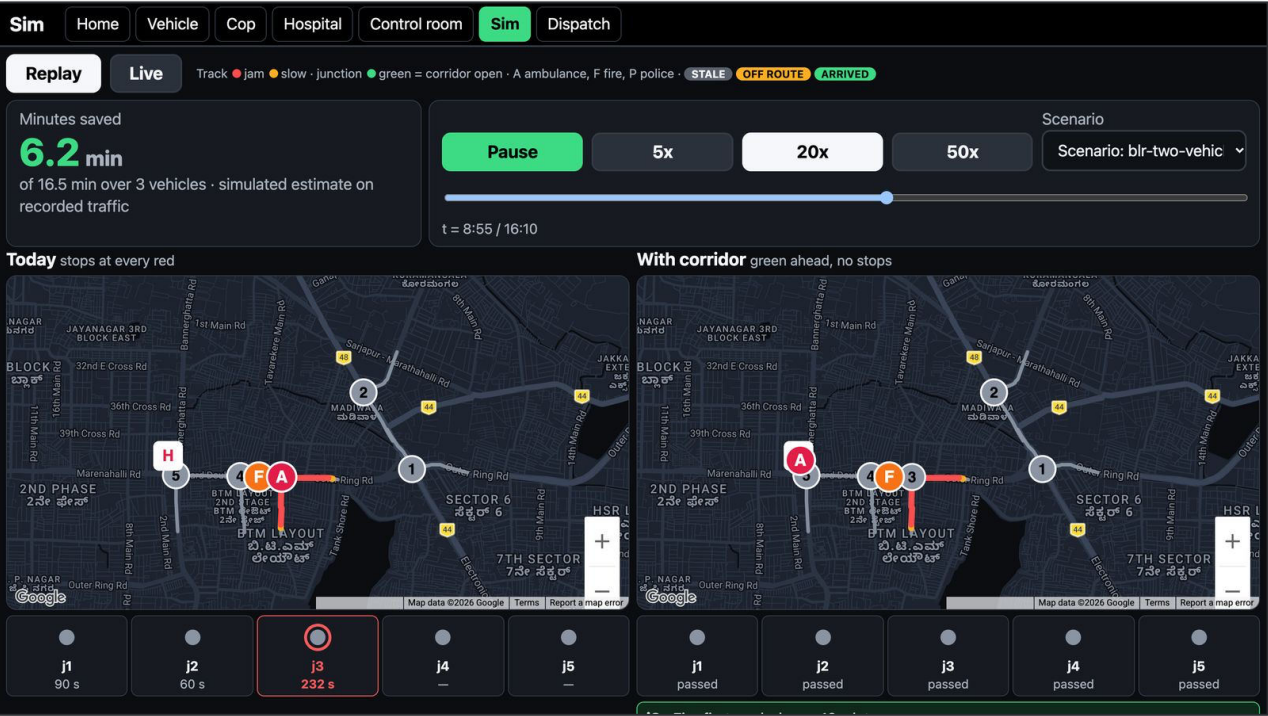
city-month at 100 runs
(illustrative)

Routes	$3/\text{min} \times 60 \times ₹0.85$	₹153
Gemini	$17 \times ₹0.02$	₹0.34
Text-to-Speech		₹1.30

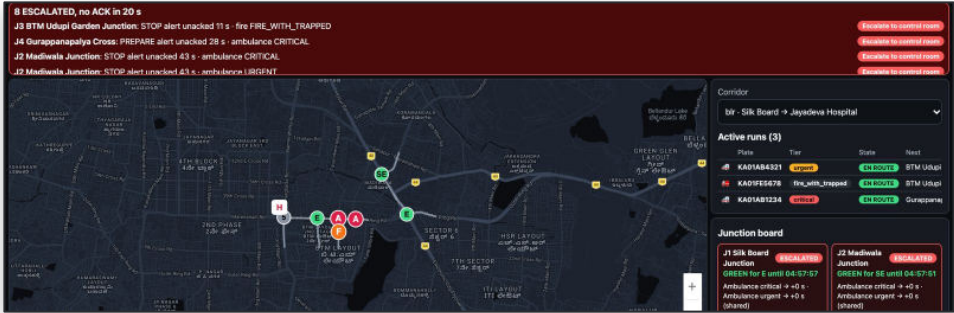
Lever: cache Routes beyond today's 20 s

Snapshots of the prototype

The loop, mid-run, from the live rehearsal



Replay simulator



Control room



STOP alert, spoken

Snapshots of the prototype (routing agent)

Live routing trace: Jayadeva chosen, two alternatives rejected

```
1 required_capabilities(critical) -> cath_lab
2 list_hospitals(blr) -> 3 hospitals
3 check_diversion(Jayadeva) -> accepting
4 check_diversion(Apollo) -> on diversion
5 eta_to(Jayadeva) -> 373 s, 2.6 km
6 Added to baseline: icu
```

Destination Jayadeva, confidence 1.0

Jayadeva Institute of Cardiovascular Sciences · 6:13

recommendation only (scenario run)

- Jayadeva Institute is the most suitable facility as it has both the required cath lab and ICU capabilities, with 4 beds available.
- It was chosen over Apollo Hospital because Apollo is currently on diversion, and Fortis does not have a cath lab.

```
1. called required_capabilities(critical) -> cath_lab
2. called list_hospitals(blr) -> 3 hospitals: blr_jayadeva, blr_apollo_bg,
   blr_fortis_bg
3. called check_diversion(Jayadeva Institute of Cardiovascular Sciences) ->
   accepting
4. called check_diversion(Apollo Hospital Bannerghatta Road) -> on diversion
5. called eta_to(Jayadeva Institute of Cardiovascular Sciences) -> 373 s, 2.6 km
6. capabilities: cath_lab (Required for immediate intervention for suspected
   STEMI/ACS.); icu (Required for patient with cardiogenic shock (hypotension,
   tachycardia) and respiratory distress.). Added to baseline: icu
```

The trace as shown in /vehicle.

Performance report / benchmarking

Synthetic-voice results, scripted-run timings

90 %

field accuracy

100 %

tier accuracy

3.5 s

mean extraction

12 synthetic clips, 11 scored for tier, 1 intervention note; not field recordings. Before the fix: 87 % / 73 %.

17 / 37 ms

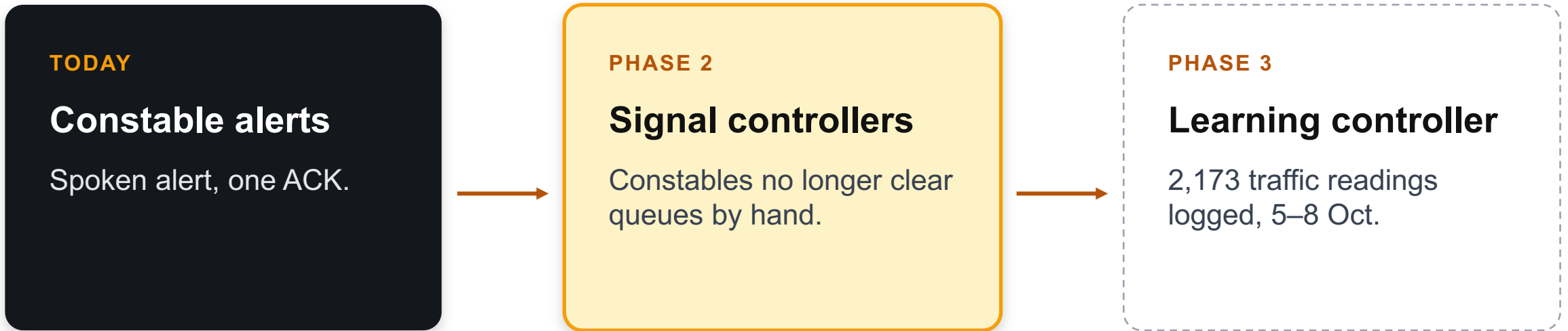
/location p50 / p95, offline

412 Python · **161** web tests

98 % coverage

Additional details / future development

From constable to controller, in three steps



Signals simulated, patients synthetic, minutes saved estimated.

Sustainability

We size the green to the queue, so cross-traffic idling is bounded by the clearance time.

0.25_L

idling fuel avoided per cleared
queue (illustration: 20 vehicles ×
60 s × 0.76 L/h)

2.35_{kg}

CO₂ per litre of petrol (EPA)

SDG 3.8 · 11.2

emergency care access ·
sustainable transport

Idle burn: Argonne National Laboratory (Gaines, Rask, Keller), US DOE, 2014. CO₂: US EPA, GHG Equivalencies Calculator, 2024. The per-queue arithmetic is an illustration, not a measurement or a net saving.

Links

Try it, read the code, watch the demo

GITHUB PUBLIC REPOSITORY



github.com/Nandish3010/ideal-disco

DEMO VIDEO (3 MINUTES)

[link + QR at
submission]

FINAL PRODUCT



green-corridor-2026.web.app

SUBMISSION DECK



green-corridor-2026.web.app/deck.pdf

Warn the constable before the siren arrives.

Try it: green-corridor-2026.web.app

Read it: github.com/Nandish3010/ideal-disco

Built by Nandish